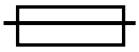
11 PLUS PREPARATION

NON-VERBAL REASONING PAPER

Name:	
Date of Birth:	
Today's Date:	

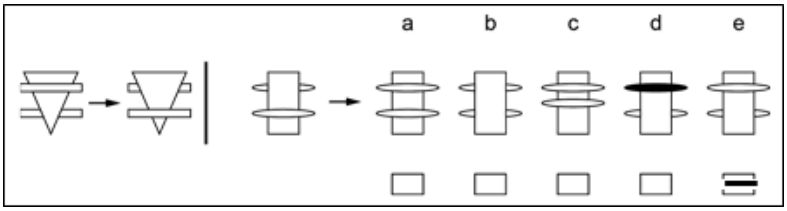
READ THESE INSTRUCTIONS CAREFULLY BEFORE STARTING.

- You have 20 minutes to complete the paper.
- Mark your answer like this: 
- This paper features a total of 33 questions from 13 different question types. An example is given at the start of each question type.
- Each question is worth one mark. If you can't answer a question, move on.
- If you finish early check your answers for mistakes.
- Once the test has started you may not ask for help.

1. LIKE SHAPES / TYPE 1

Work out which of the answer shapes is related to the question shape in the same way that the example shapes are related.

Example



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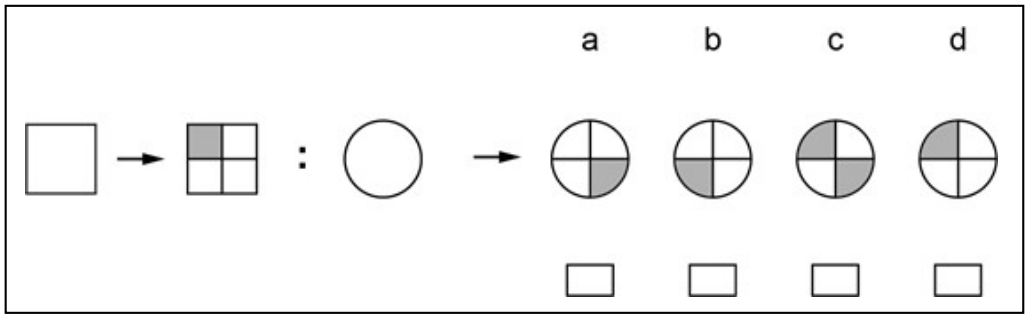

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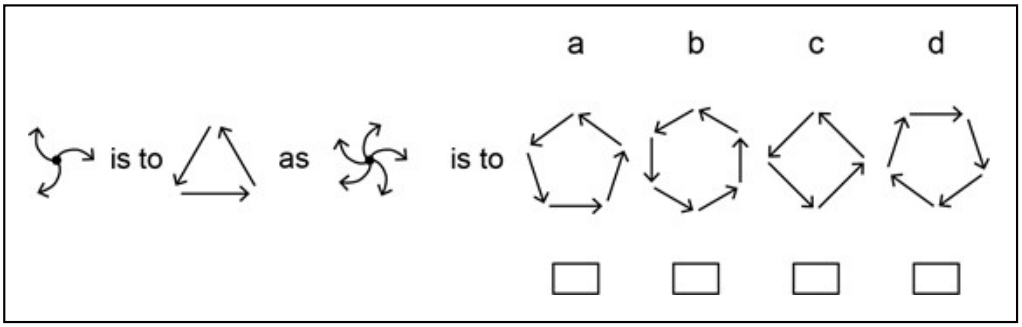

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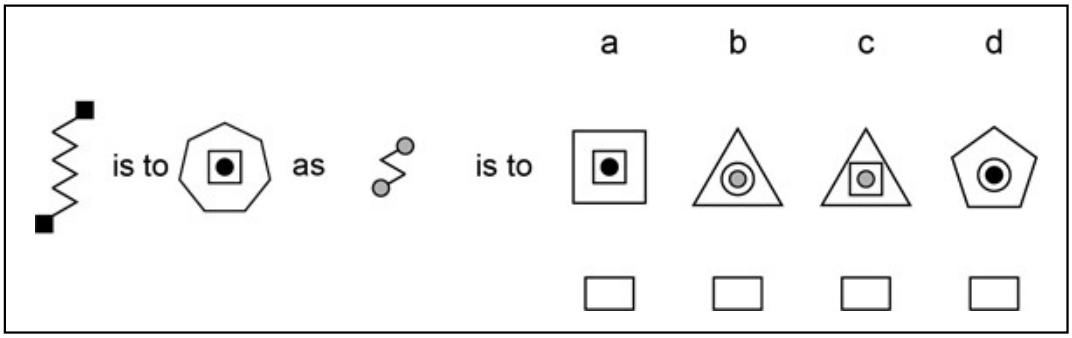
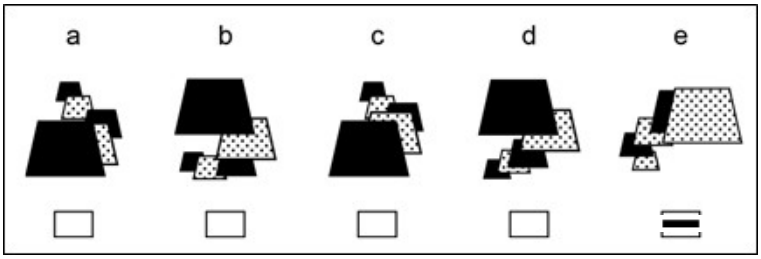


2. ODD ONE OUT / TYPE 2

Work out which of the shapes is different to the rest.

Example



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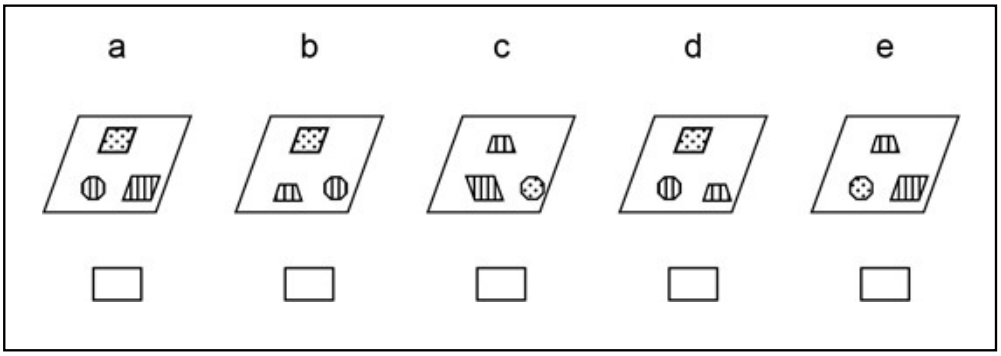

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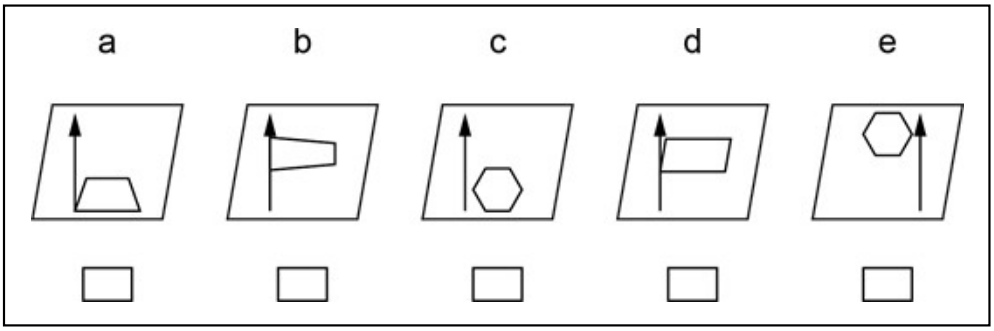

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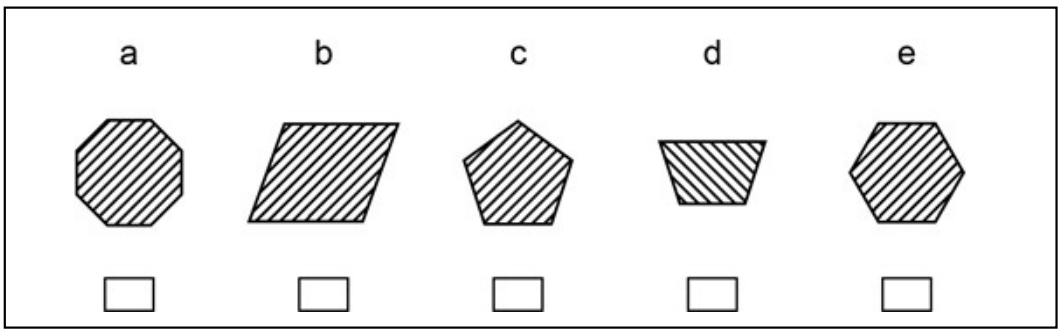
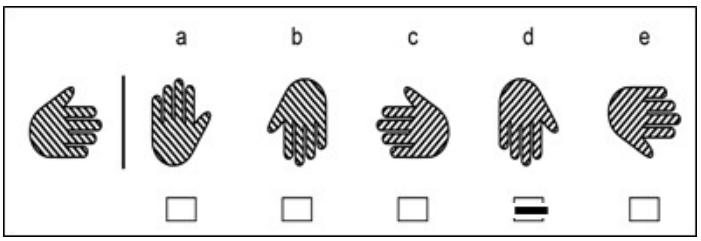


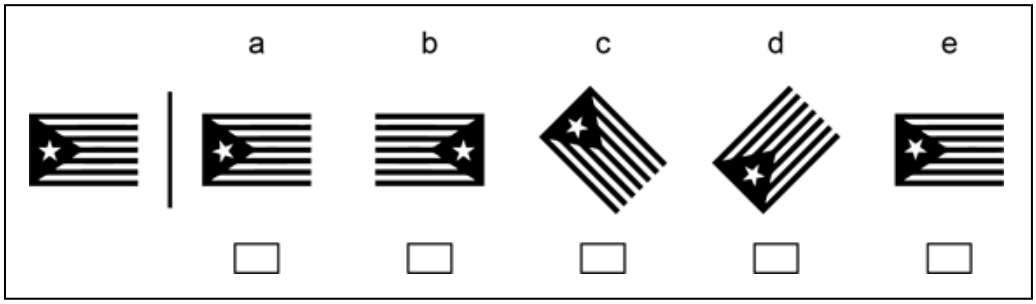
3. ROTATIONS / TYPE 3

Work out which of the answers is a rotation of the question shape.

Example



1



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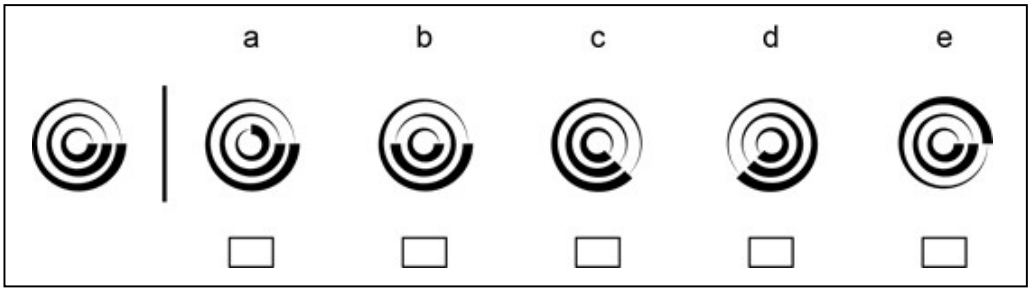

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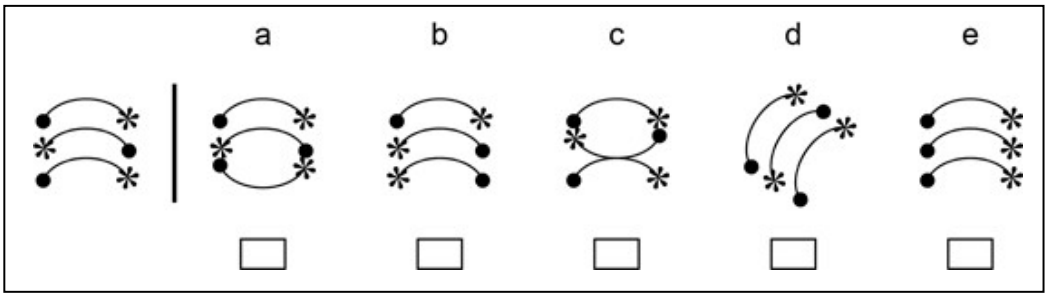
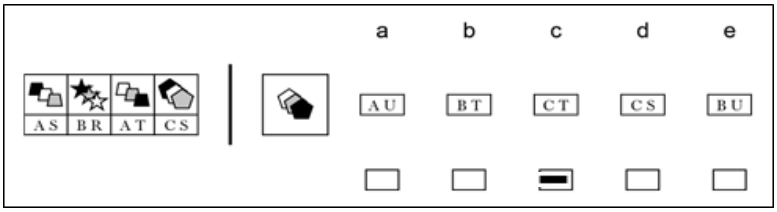


4. CODE BREAKER / TYPE 4

Look at the sequence to work out how the codes match the shapes, then select the correct answer code to match the question shape.

Example



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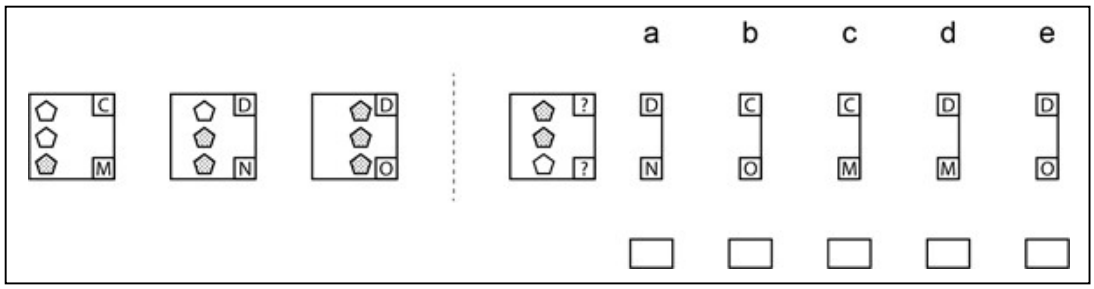

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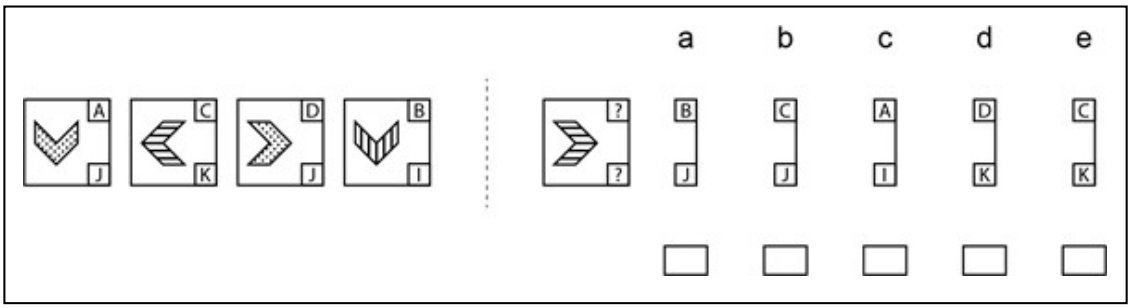

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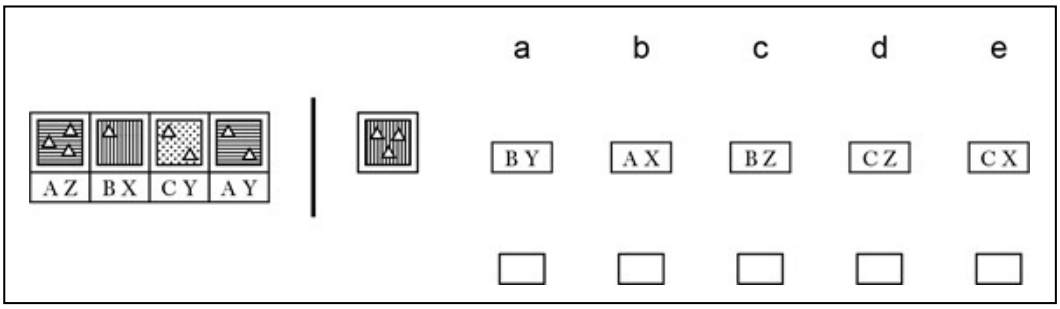
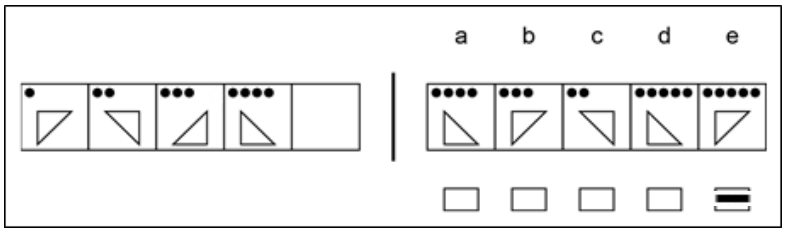


5. MISSING SEQUENCE / TYPE 5

Select the answer that correctly completes the sequence.

Example



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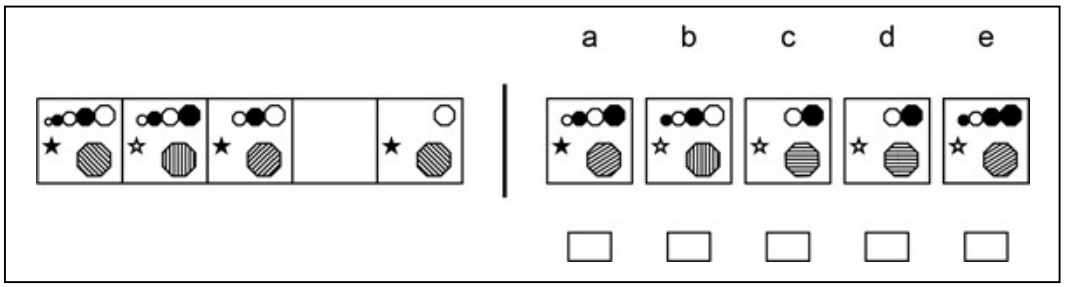

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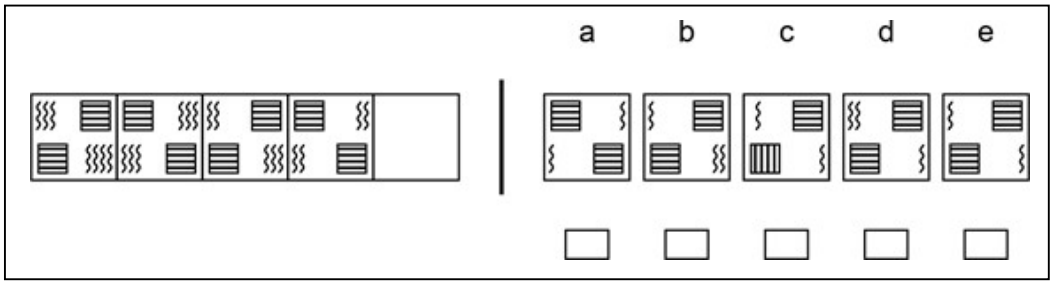

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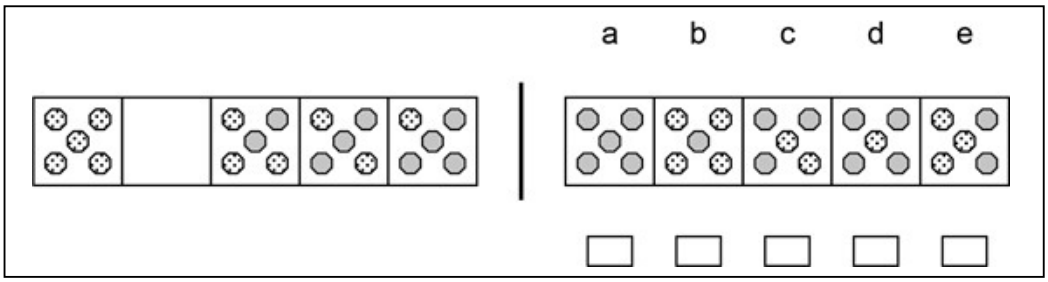
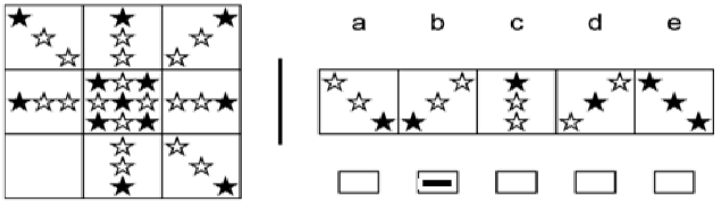


6. MISSING SQUARE / TYPE 6

Select the answer that correctly completes the grid.

Example



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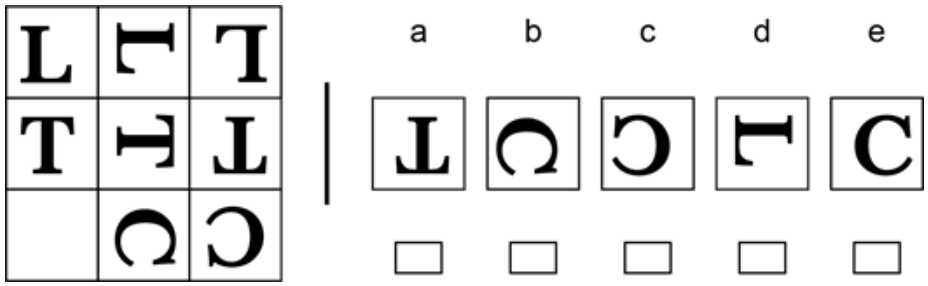

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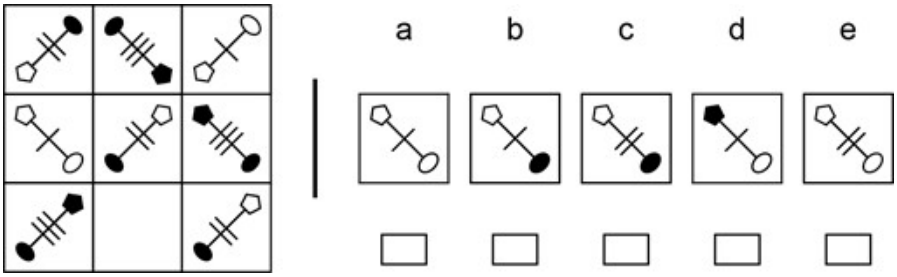

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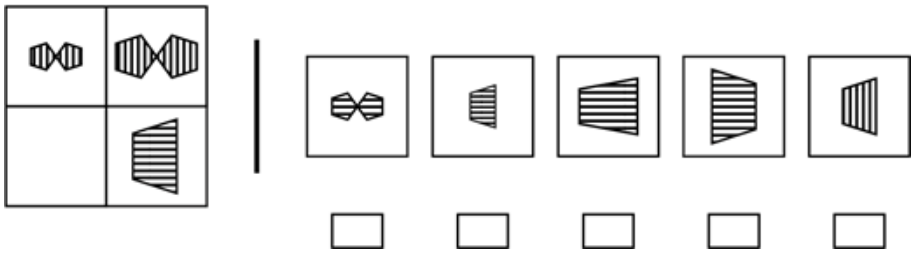
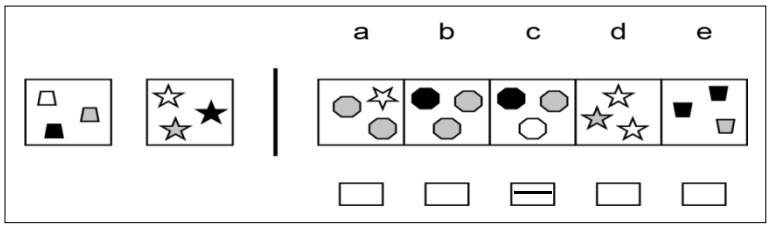


7. MOST LIKE / TYPE 7

Select the answer that is most similar to the two given shapes.

Example



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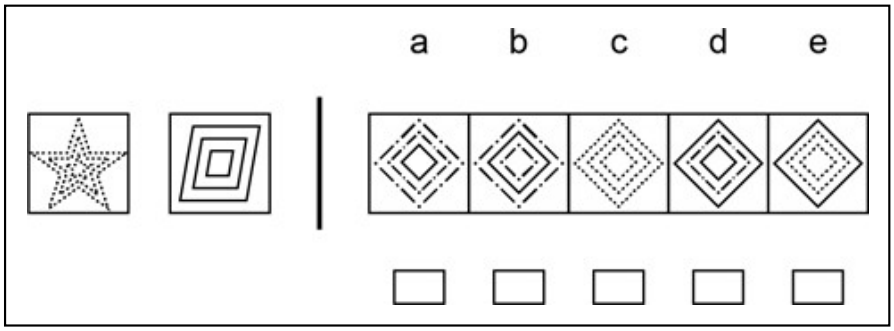

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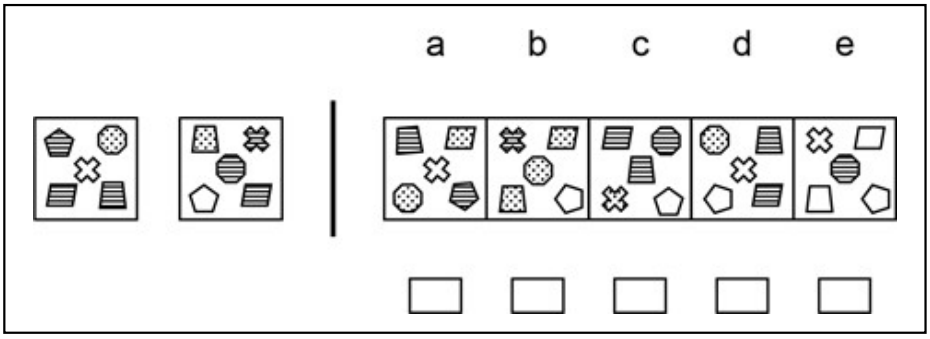

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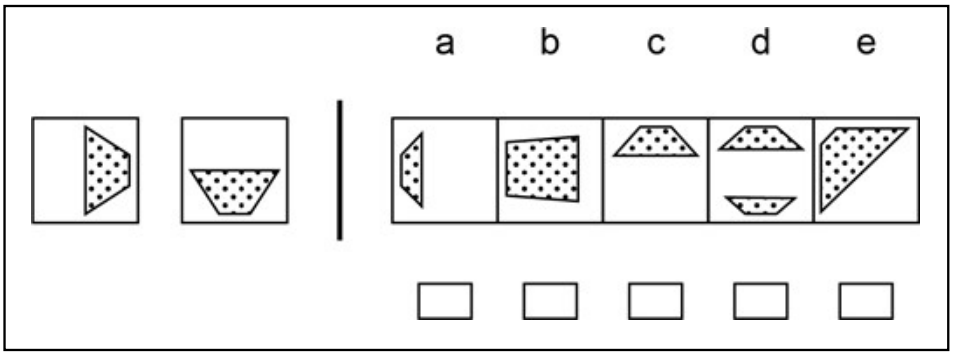
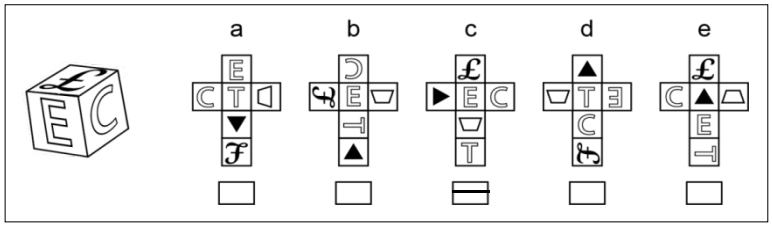


8. NETS / TYPE 8

Select the cube that is made from the given net, or net that makes the given cube.

Example



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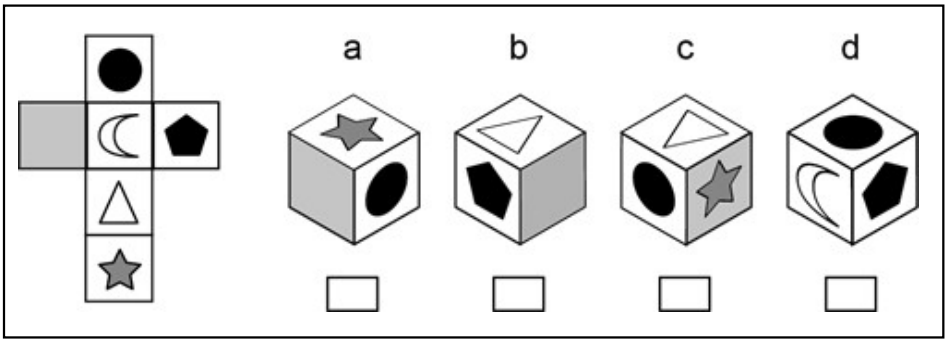

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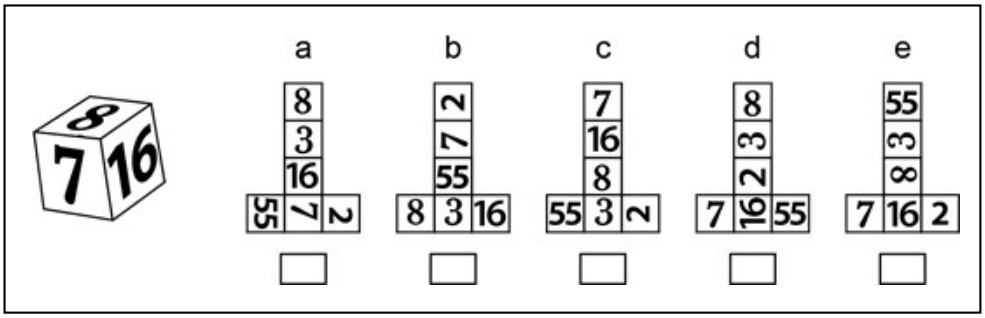

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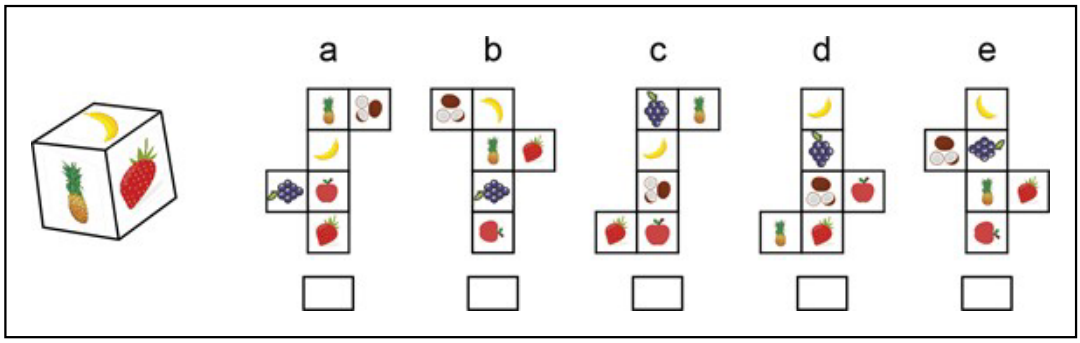
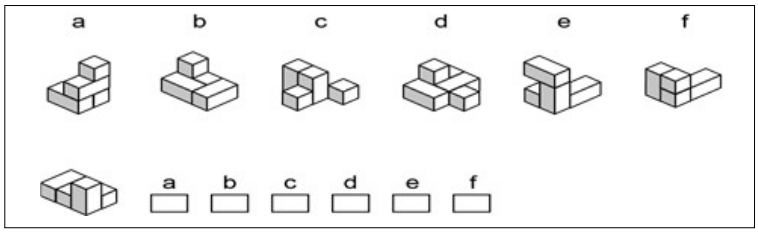


9. 3D SHAPES / TYPE 9

You will need to do one of the following:
Select the answer that is a rotation of the given 3D shape; or
Select the building blocks that make the given 3D shape; or
Select the correct 2D view of the given 3D shape.

Example



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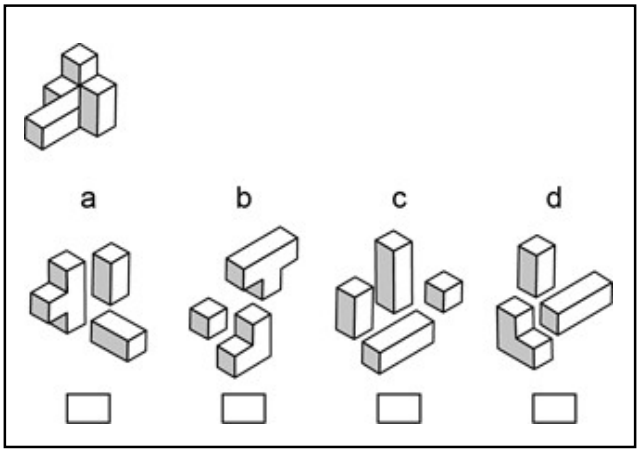

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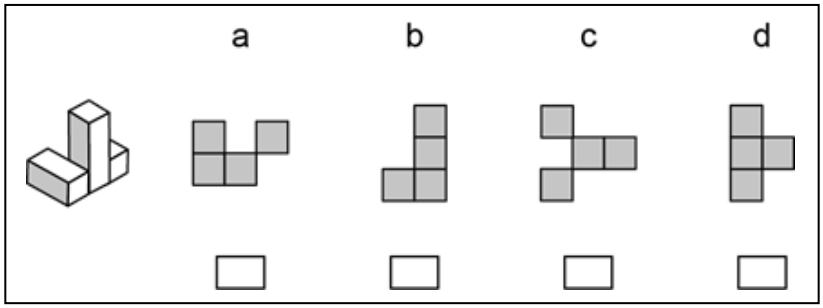

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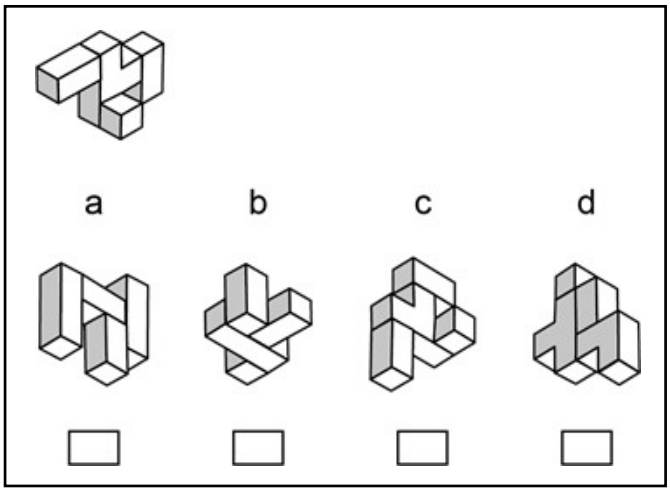
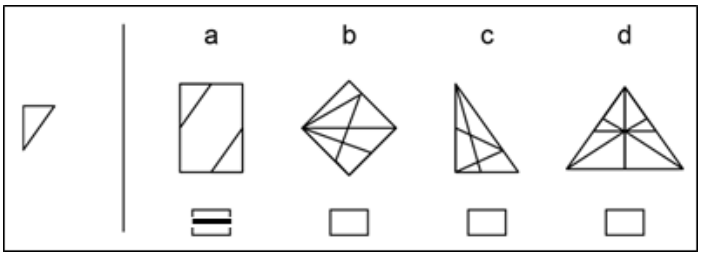


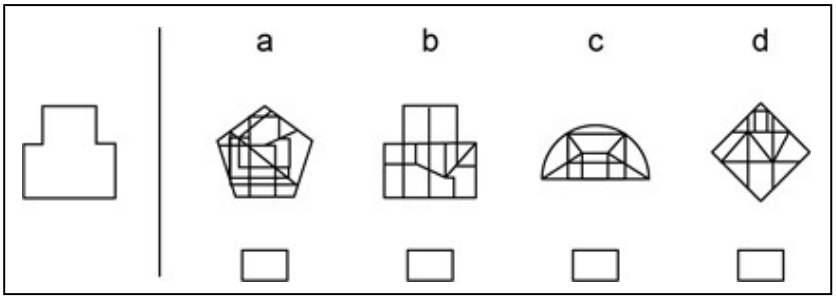
10. HIDDEN SHAPE / TYPE 10

Find the answer option that contains the shape hidden in exactly the same size and orientation.

Example



1



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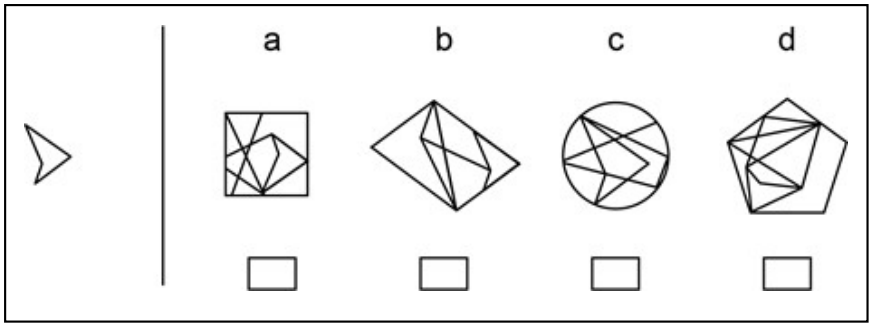

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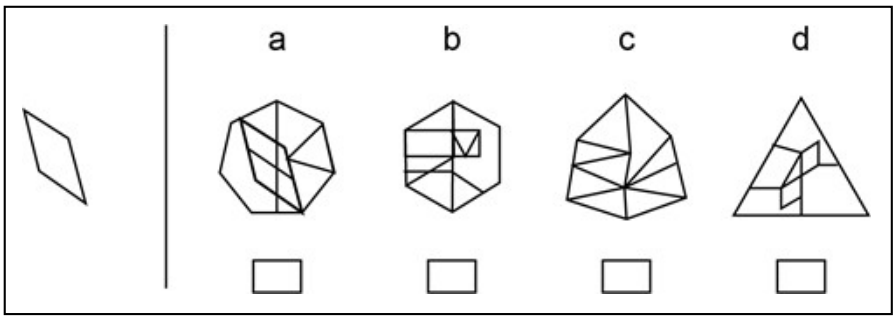
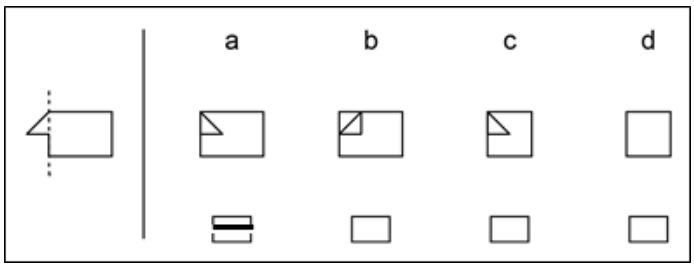


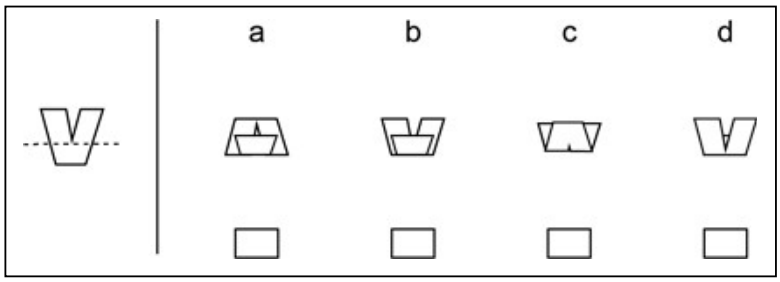
11. FOLD ALONG THE DOTTED LINE / TYPE 11

Select the option that will be formed when the shape is folded along the dotted line.

Example



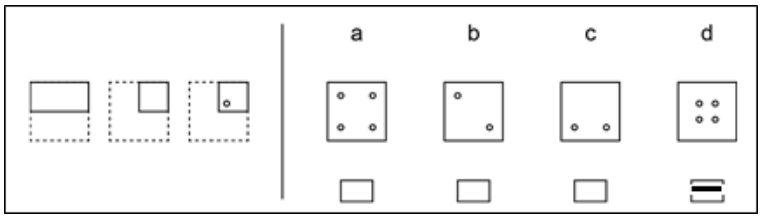
1



12. HOLE PUNCH / TYPE 12

The sequence shows a shape being folded and then hole punched. Select the option that will be created when the hole punched shape is unfolded.

Example



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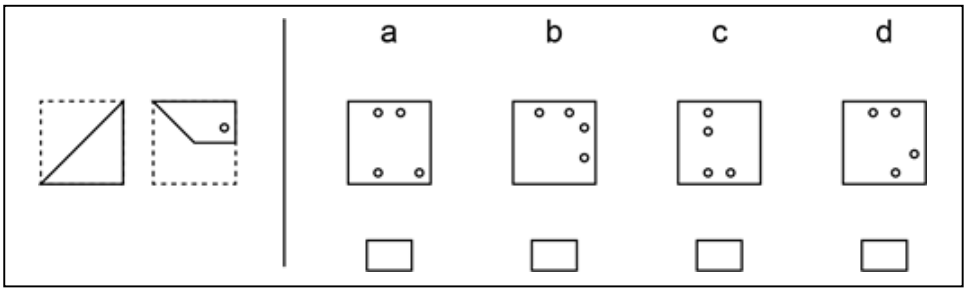
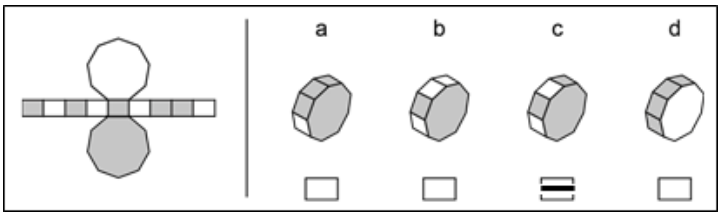


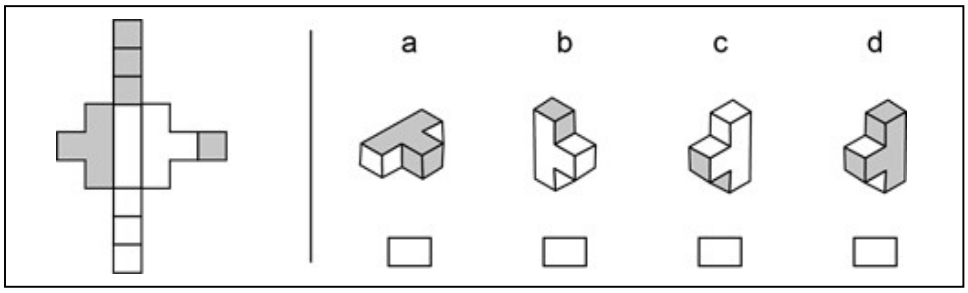
13. SHADED NETS / TYPE 13

Select the 3D shape that is created from the shaded net.

Example



1



Non-Verbal Reasoning

Like Shapes

1. **D** - (The shape becomes a grid with the top left portion shaded.)
2. **A** - (Arrows pointing clockwise from a point become a shape made from arrows pointing anti-clockwise. Therefore, 5 arrows pointing clockwise from a point will become a pentagon of arrows pointing anti-clockwise.)
3. **B** - (The number of lines between the shapes denotes the number of sides that the new outer shape will have. The shape at either end of the lines denotes the shape that will be the new middle shape. The colour of the shape at either end of the lines denotes the colour of the circle that will be the new innermost shape.)

Odd One Out

1. **A** - (The only answer without 3 different shapes within the rhombus is A which has two rhombi and an octagon.)
2. **E** - (The arrows are all on the left hand side of the squares except for in E where it is on the right hand side.)
3. **D** - (The diagonal lines in D are going in the opposite direction to the diagonal lines in all the other shapes.)

Rotations

1. **D** - (D is a rotation of around 45 degrees anti-clockwise of the original.)
2. **C** - (C is a rotation of around 45 degrees clockwise of the original.)
3. **D** - (D is a rotation of around 45 degrees anti clockwise of the original.)

Code Breaker

1. **A** - (This is a very tricky question. You can work out that D stands for the middle square being shaded as there two of them and it is the only similarity. Therefore, this means that C stands for the middle square not being shaded. All of the bottom letters are different and there are two differences between all of them. Firstly, the three squares have a different number of stars shaded. So, M is one, N is two and O is three. Or you could say M is the stars being on the left hand side, N in the middle and O on the right. Whichever of the two that you use, the answer is B, for the middle square

being shaded, and N for there being two stars shaded and being in the centre of the square. So, the final answer is DN.)

2. **D** - (You can work out that J stands for the shading of the chevron being dotted as there are two of them. Therefore, K stands for the shading being horizontal lines and I stands for the shading being vertical lines. All of the top letters are different and the only difference between all four of them is the direction that the chevrons are pointing in. So, the chevron pointing to the right must be D. So, the final answer is DK.)
3. **C** - (You can work out that A stands for there being horizontal lines in the square as there are two of them. Therefore, B stands for there being vertical lines and C stands for dotted shading. In the right hand letters there are two Ys and the similarity between the two are that there are two triangles in each. So, three triangles in the square must be Z. So, the final answer is BZ.)

Missing Sequence

1. **C** - (There are three changes to each square. Moving from left to right; the lines in the circle rotate 45 degrees clockwise, the star changes from black to white or white to black and the smallest octagon at the top disappears and the remaining octagons change colour from black to white or from white to black. So, the final answer must have two octagons with white on the left and black on the right, a white star and horizontal lines in the circle so must be C.)
2. **B** - (As you move from left to right two changes happen each time; the small squares alternate between left and right and the squares alternate between removing one line from the bottom and one line from the top. In the missing square one line should be taken from the top compared to the previous square and the top small square should be on the right, so the answer is B.)
3. **B** - (As you move from left to right one of the octagons with dotted shading changes to a lightly shaded octagon. This means that in the second square there should be four octagons with dotted shading and one lightly shaded, so the answer is B.)

Missing Square

1. **E** - (In each row, as you move from left to right the letters are rotated 90 degrees clockwise. So the missing square should be a rotation of 90 degrees anti-clockwise of the middle square in the bottom row. The answer is therefore E.)
2. **A** - (As you move across the two completed rows from left to right, a number of things happen: the direction of the line alternates between left to right and right to left, the number of the small dashes on the line increases from 1 to 2 to 3 and then back to 1 etc. and the colours of the shapes on the end changes from one white and one black to both black to both white and then back to one white and one black. So the missing square should have the line pointing from the bottom right to the top left, it should have 1 dash on the line and the shapes on the end should both be white, so the answer is A.)
3. **B** - (From the top row you can see the only difference is that the shape from the left square has got smaller. By applying this to the bottom row you can work out that the missing square is B.)

Most Like

1. **C** - (The three shapes in both the squares on the left have the same type of outlines- the left square has three shapes with a dashed outline and the right square has three shapes with a solid line. The only answer with three shapes with the same outline is C as it has three shapes with dashed outlines.)
2. **C** - (In the two squares on the left there are three shapes with horizontal lines, one white and one with dotted shading in each. The only answer that has the same is C.)
3. **E** - (In each of the squares on the left there is one large, dotted trapezium that is the size of about half of the square. The trapeziums in A and C are both too small, the trapezium in B is too big and there are two trapeziums in D, so the answer has to be E as it is the only dotted trapezium that is the size of about half the square.)

Nets

1. **D** - (You can see from the net that the black circle must be above the crescent moon. The black pentagon will fold across to make an edge with the black circle and crescent moon.)

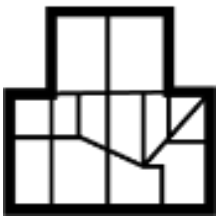
2. **E** - (You can see from the cube that the number 7 and 16 must appear adjacent to one another upright. This is only possible in answer E. The number 8, apparent horizontally above 16, will then fold around to appear upright above 7.)
3. **B** - (You can see from the cube that the banana must appear above the pineapple. This is only possible in net B. The strawberry is next to the pineapple in the correct orientation and can fold around to form an edge with the banana.)

3D Shapes

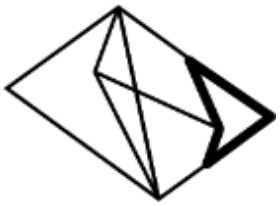
1. **D** - (The shape must be made from one shorter cuboid, one longer cuboid, and one 'L' shaped block.)
2. **B** - (Viewed from above, the shape will appear as a reverse 'L' shape.)
3. **C** - (C is a rotation of approximately 90 degrees anti-clockwise in the y-axis and 90 degrees anti-clockwise in the z-axis)

Hidden Shape

1. **B** -



2. **B** -



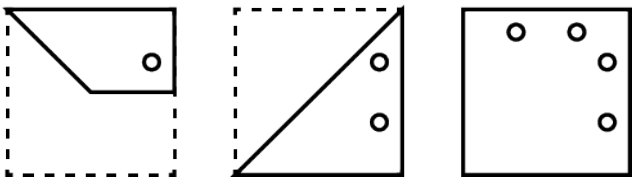
3. **A** -

Fold Along the Dotted Line

1. **D** - (The bottom portion folds behind the top portion so that a small part of the bottom portion is visible in the background.)

Hole Punch

1. B -



Shaded Nets

1. **C** - (When the net is folded down and the resulting shape is stoop on its end and viewed from right, the white 'T' shape is visible with grey shaded squares at the bottom.)